YEAR 7 · UNIT 4 REVIEW · SCIENCE UNIT 3 · HOMEWORK 10

Faster in January

Part 1 — Mathematics Unit 4 again, with new numbers
Part 2 — Science 3.2, 3.3 and the football model, in words and in decimals

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 30 October 2026

How to do this homework

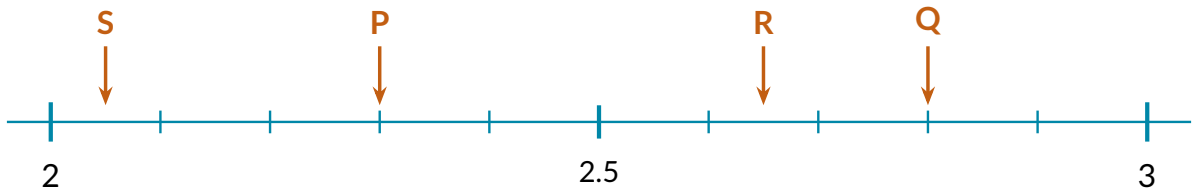
Six sections. A to C are **Unit 4 again**, with new numbers. D to F are this week's science, and **F does it in decimals**.

- **Write the calculation before you write the answer. Estimate first**: the estimate tells you where the point goes.
- **No calculator.** In D, E and F, answer in **full English sentences** where it says so. *"Gravity"* is not a sentence. *"Gravity pulls the planet towards the Sun"* is.

Section A · Unit 4 Review — Comparing, Adding and Subtracting

A1 Read the line

The line goes from **2** to **3**. Every small line is **0.1**. **(a)** Write the number each arrow points to.



P _____	Q _____	R _____	S _____

(b) Write the four numbers in order, **starting with the largest** (the biggest).

_____

A2 Which is bigger?

Write <, > or = in each box. Look at the whole-number parts first. Then **fill with zeros**: 0.5 = 0.50.

(a) 0.5 ☐ 0.45

(b) 2.1 ☐ 2.09

(c) 8.08 ☐ 8.8

(d) 6.4 ☐ 6.400

(e) 15.6 ☐ 9.87

(f) 1.25 ☐ 1.3

A3 Line up the points

Written method

Write the numbers with the **points in a line**. Fill the gaps with **zeros**. The point in the answer goes **straight down**.

(a) 7.8 + 4.56

(b) 23.4 + 0.87 + 5

(c) 16.25 + 3.9

(d) 10 − 3.42

(e) 6.3 − 2.75

(f) 0.9 − 0.36

Section B · Unit 4 Review — Multiplying and Dividing

Where the point goes

× : take the point out and multiply, then **estimate** to put it back. 3.8 × 12: 38 × 12 = 456, and 4 × 12 = 48, so **45.6**.
÷ : short division. The point goes **straight up**. If a digit is too small to divide, write **0**. To divide by a 2-digit number, write its **times table** first.

B1 Point out, point back

Fill in the table. The first one is done for you.

Calculation	Estimate	Without the point	Answer
2.4 × 13	2 × 13 = 26	24 × 13 = 312	31.2
0.6 × 9			
3.45 × 6			
7.2 × 15			
2.8 × 23			

B2 The point goes straight up

Use short division. For **(e)** and **(f)**, write the times table first.

(a) 9.45 ÷ 5

(b) 4.32 ÷ 4

(c) 0.72 ÷ 9

(d) 7.3 ÷ 2

(e) 52.5 ÷ 15

(f) 81.6 ÷ 24

Section C · Unit 4 Review — Words, Quick Ways and Money

C1 From English to a calculation

Word help — the words that choose the operation

each	for one of them. *3 pens at 2 dollars each*: 3 × 2
altogether	all of it, the total
twice as far	2 times as far
what must you add to 4 to make 7?	7 − 4. It says *add*, but you **subtract**
3 less than 10	10 − 3. The **second** number comes **first**

Write the **calculation first**, then the answer. Give units.

(a) What must you add to 4.65 to make 7?

Calculation: _____ **Answer:** _____

(b) A ribbon 2.4 m long is cut into 16 equal pieces. How long is each piece?

Calculation: _____ **Answer:** _____

(c) Mr Bowen walks 2.5 km on Monday. On Tuesday he walks twice as far. How far does he walk altogether?

Calculation: _____ **Answer:** _____

(d) What is 3.5 less than 10?

Calculation: _____ **Answer:** _____

C2 Use a quick way

Quick ways

× 0.5 = **half of** × 0.25 = **a quarter of** × 0.07 = ÷ 100, then × 7
× 9: × 10, then **take away** one lot. × 11: × 10, then **add** one lot.
Partitioning: split the number. 4.3 × 6 = 24 + 1.8

Write the quick way, then the answer.

(a) 4.7 × 9 = _____ = _____

(b) 6.2 × 11 = _____ = _____

(c) 5.3 × 4 = _____ = _____

(d) 64 × 0.25 = _____ = _____

(e) 300 × 0.07 = _____ = _____

(f) 9.6 × 0.5 = _____ = _____

C3 Change from 10 dollars

Money has **2 decimal places**: 7.8 dollars is written 7.80 dollars.

Mr Bowen buys 3 bánh mì at 1.35 dollars each and 2 iced coffees at 1.90 dollars each. He pays with a 10-dollar note. How much change does he get?

C4 Mr Bowen's homework

Only **one** of these is right. Find it, and correct the other three.

(a) 4.5 + 2.7 = 6.12 _____

(b) 9.6 ÷ 3 = 3.2 _____

(c) 3.40 > 3.4 _____

(d) 60 × 0.4 = 2.4 _____

C5 Now you write the question

The rules

A **real** situation, in full sentences, ending in a **question**. Use a **word** from C1, give **units**, and come out at the answer given.

Write a word problem for $12 \times 2.40 = 28.80$

Section D · Science 3.2 — How the Solar System Formed

D1 Key words

Write the letter of the right meaning next to each word.

nebula		**A**	an idea that can be tested
hypothesis		**B**	one flat surface
evidence		**C**	a cloud of dust and gas in space
model		**D**	facts from observations or experiments
limitation		**E**	something a model cannot do, gets wrong or leaves out
plane		**F**	a way of showing something that is hard to observe directly

D2 Support or contradict?

The **hypothesis**: **The Sun and the planets formed from one spinning cloud of dust and gas.**

Does each fact **support** the hypothesis, or **contradict** it? Tick one box.	**supports**	**contradicts**
All the planets orbit the Sun in the same direction.		
All the orbits are in the same plane.		
Venus spins the opposite way to most planets.		
HL Tauri is a young star with a flat disc of dust round it.		
Uranus spins on its side.		
Most moons orbit their planet in the same direction.		

(b) Two facts contradict the hypothesis. So is it wrong? Answer in a full sentence. Use the words **most of the evidence**.

D3 From a cloud to the Sun and planets

(a) Put the four steps in order. Write **1**, **2**, **3** and **4** in the boxes.

☐ The hot middle becomes the Sun.

☐ Gravity pulls the nebula together. It spins faster.

☐ Dust sticks together and grows into planets.

☐ The nebula flattens into a spinning disc.

(b) Why do all the planets orbit the Sun in the same direction? Answer in a full sentence. Start: **"They all formed from …"**

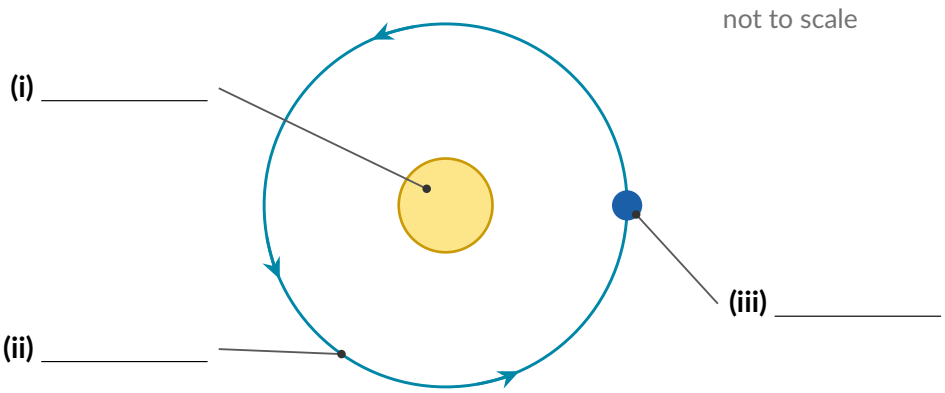
(c) Mercury, Venus, the Earth and Mars are small and rocky. Why? Start: **"Near the Sun, it was too hot for …"**

(d) The Sun has **99.8%** of all the mass in the Solar System. % means *out of 100*. How many % are left for everything else? _____

Section E · Science 3.3 — Movement in Space

E1 What keeps a planet in orbit?

(a) Label the drawing. Use these words: **Sun** · **orbit** · **planet**



(b) Draw an **arrow** on the planet to show the pull of the Sun's **gravity**.

(c) The arrows show which way the planet moves. Now the Sun's gravity **switches off**. Draw a **dashed line** to show where the planet goes.

E2 Which forces?

Which forces act on each one? Tick **every** box that is true.	**gravity**	**air resistance**
The Earth, orbiting the Sun		
The Moon, orbiting the Earth		
A ball falling from the balcony		
A plane landing on a runway		

(b) Why do the planets never slow down? Start: **"Space is a …, so …"**

Section F · Science in Decimals — Faster Near the Sun

Speed, from Science 3.3

Speed: how fast something moves. **km/s** = kilometres **per** second.
As the distance from the Sun **increases**, the speed of the planets **decreases**. The Sun's pull is **weaker** far away.

F1 How much faster? How many times as fast?

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
speed (km/s)	**48**	**35**	**30**	**24**	**13**	**10**	**7**	**5**

Word help — two questions that look the same

how much faster? the difference. **Subtract**

how many times as fast? **divide** the bigger speed by the smaller one

Fill in the table. The first row is done for you.	**How much faster?**	**How many times as fast?**
Mercury and Neptune	48 − 5 = 43 km/s	48 ÷ 5 = 9.6
Mercury and the Earth		
Mars and Saturn		
The Earth and Mars		

F2 The Sun is a football

In the lesson the Sun became a football, and everything else shrank by the same amount.

	the Moon	**the Earth**	**Neptune**	**Jupiter**	**the Sun**
Model width	0.5 mm	2 mm	7.8 mm	2.3 cm	22 cm
Like	a grain of sand	a pinhead	a pea	a grape	a football

(a) Write the five model widths in order of size, starting with the **smallest**. Remember: **1 cm = 10 mm**.

(b) In the model, the Earth is **24 m** from the football. Mr Bowen builds the model in his classroom, which is **8.5 m** long. He puts the football against one wall. How far past the other wall must the Earth go?

Calculation: _____ **Answer:** _____

(c) Give **one limitation** of the football model. Start: **"One limitation of the football model is …"**

F3 Faster in January

The Earth's orbit is *almost* circular. In January the Earth is a little **closer** to the Sun than in July.

	Distance from the Sun	Speed
January	147.1 million km	30.3 km/s
July	152.1 million km	29.3 km/s

(a) How much closer to the Sun is the Earth in January?

Calculation: _____ **Answer:** _____

(b) In which month is the Earth faster? Why? Use the words **closer** and **pull**. Start: **"The Earth is faster in …, because …"**

(c) How much faster is the Earth in January than in July?

Calculation: _____ **Answer:** _____

(d) Find the **mean** of the two speeds.

Calculation: _____ **Answer:** _____

(e) Round your mean to the nearest whole number. _____

In the lesson, the Earth's speed was **30 km/s**. What do you notice?